

Description	Owners/Teams	Impact	Effort (FTE)	Milestones (M) and Decision Points (D) (year: description)	Risks and External Dependencies	Prob. of Applicability On/Success/Adoption (Low, Medium, High)
1.3 RAW event reduction - tracker	Development Team: CMS CMS ownership: CMP task (Ivan Amos Cali)	Factor 2 reduction (whole HI-MinBias) RAW event 15% reduction (whole pp event)	0.5 0.5	'22 M: current HI raw event reduction 30-40% (compression), 40% (squared clusters) with 3% tracking efficiency loss at high-pT '22 D: pp acceptable physics performance for a given reduction '23 M: implementation of reduced pp raw event size	Risks: physics performance not adequate after size reduction	Applicability to pp: Medium
2.35 AOD, Mini-AOD, Nano-AOD reduction in ROOT RNTuple	Development team: community-wide Root collaboration CMS ownership: Core Software (Matti)	Reduce storage by 5-15 %, reduce decompression CPU cost by 20 %.	3	'23 M: NanoAOD, MiniAOD demonstrators/benchmarks '24: Persistent EDM and benchmarks	Dependencies: Evolution in ROOT, workflow production, computing model Risks: <u>On-time</u> delivery of production quality RNTuple by ROOT team, I/O expertise in ROOT and CMS	Adoption: High
	Kortelainen, Shahzad Muzaffar + Ph. Canal?)					
3.8 Partial Prompt Reconstruction	CMS ownership: Computing Operations(C. Paus, A. Wightman), L1s, David Lange	~5% tape, ~10% disk savings	0.2 0.1 0.1	'18 M: 10B b decays recorded '22 M: B parking at 3kHz '23-'25 M: exercise parking campaigns and associated tape recall '32 M: "Emergency brake"	Risks: Delays in time to publication of physics papers	Adoption: hopefully Low (emergency brake in case other options fail)